

WORK EXPERIENCE

Harbin Institute of Technology

PostDoc, Associate Researcher

- Cooperative Advisor: Prof. Zongquan Deng

Harbin, Heilongjiang

Jan. 2025 – Present

EDUCATION

Eidgenössische Technische Hochschule Zürich (ETH Zürich)

Visiting Ph.D. of Robotics

- Research topic: Physics-aware navigation for legged robots
- Advisor: Prof. Marco Hutter

Switzerland, Zürich

Jul. 2022 – Jan. 2024

Harbin Institute of Technology

Ph.D. of Aerospace Science and Technology

- Research Interests: Space robots, Terrain sensing, Physical scene understanding
- Advisor: Prof. Liang Ding

Harbin, Heilongjiang

Sep. 2019 – Dec. 2024

Harbin Institute of Technology

Master of Mechanical Engineering

- Dissertation: “Research on Modeling of Terrain Geometrical and Mechanical Properties Based on Planetary Rovers’ Visual Information”
- Honors: First-class scholarship of Harbin Institute of Technology

Harbin, Heilongjiang

Sep. 2017 – Jul. 2019

Hunan Normal University

Bachelor of Mechanical Design, Manufacturing and Automation

- Honors: Excellent Graduation Thesis, National scholarship, National motivational scholarship

Changsha, Hunan

Sep. 2013 – Jun. 2017

PUBLICATIONS

- S. Cheng, R. Zhou, W. Feng, H. Yang, H. Gao, Z. Deng, L. Ding*, **VLM-Empowered Multi-Mode System for Efficient and Safe Planetary Navigation**, *arXiv*, 2025.
- H. Wang, R. Zhou, L. Ding*, T. Liu, Z. Zhang, P. Xu, H. Gao, Z. Deng, **Whole-Body Constrained Learning for Legged Locomotion via Hierarchical Optimization**, *IEEE Robotics and Automation Letters*, **7(10)**, 7460 - 7467 (2025).
- J. Chen, J. Frey*, R. Zhou, T. Miki, G. Martius, M. Hutter, **Identifying terrain physical parameters from vision-towards physical-parameter-aware locomotion and navigation**, *IEEE Robotics and Automation Letters*, **9(11)**, 9279 - 9286 (2024).
- L. Ding*[†], R. Zhou[†], T. Yu[†], H. Yang, X. He, *et al.* **Lunar rock investigation and tri-aspect characterization of lunar farside regolith by a digital twin**, *Nature Communications*, **15(1)**, 2098 (2024).
- P. Arm*[†], G. Waibel[†], J. Preisig, T. Tuna, R. Zhou, *et al.* **Scientific exploration of challenging planetary analog environments with a team of legged robots**, *Science Robotics*, **8(80)**, eade9548 (2023).
- W. Feng, L. Ding*, R. Zhou, C. Xu, H. Yang, *et al.* **Learning-Based End-to-End Navigation for Planetary Rovers Considering Non-Geometric Hazards**, *IEEE Robotics and Automation Letters*, **8(7)**, 4084 - 4091 (2023).
- P. Xu, L. Ding*, Z. Li, H. Yang, Z. Wang, H. Gao, R. Zhou, Y. Su, Z. Deng, Y. Huang, **Learning physical characteristics like animals for legged robots**, *National Science Review*, **10(5)**, nwad045 (2023).
- R. Zhou, W. Feng, L. Ding*, H. Yang, H. Gao, *et al.* **MarsSim: A high-fidelity physical and visual simulation for Mars rovers**, *IEEE Transactions on Aerospace and Electronic Systems*, **59(2)**, 1879 - 1892 (2022).
- L. Ding*, P. Xu, Z. Li, R. Zhou, H. Gao, *et al.* **Pressing and rubbing: physics-informed features facilitate haptic terrain classification for legged robots**, *IEEE Robotics and Automation Letters*, **7(3)**, 5990 - 5997 (2022).

- L. Ding^{*†}, R. Zhou[†], T. Yu[†], H. Gao^{*}, H. Yang^{*}, *et al.* **Surface Characteristics of the Zhurong Mars Rover Traverse at Utopia Planitia**, *Nature Geoscience*, **15(3)**, 171 - 176 (2022).
- L. Ding^{*†}, R. Zhou[†], Y. Yuan[†], H. Yang, J. Li, *et al.* **A 2-year locomotive exploration and scientific investigation of the lunar farside by the Yutu-2 rover**, *Science Robotics*, **7(63)**, abj6660 (2022).
- Z. Gong, L. Ding^{*}, H. Xing, H. Gao, P. Xu, R. Zhou, Y. Lu, H. Yue, **Suppression in any configuration: A versatile coupling improved multi-objective manipulation framework for modular active vibration isolation system**, *Mechanical Systems and Signal Processing*, **166**, 108478 (2022).
- P. Xu, L. Ding^{*}, Z. Wang, H. Gao, R. Zhou, *et al.* **Contact Sequence Planning for Hexapod Robots in Sparse Foothold Environment Based on Monte-Carlo Tree**, *IEEE Robotics and Automation Letters*, **7(2)**, 826 - 833 (2021).
- P. Xu, L. Ding^{*}, H. Gao, R. Zhou, N. Li, Z. Deng, **Environmental Characterization and Path Planning for Legged Robots Considering Foot-terrain Interaction**, *Journal of Mechanical Engineering*, **56(23)**, 21 - 33 (2020). (in Chinese)
- R. Zhou, L. Ding^{*}, H. Gao, W. Feng, *et al.* **Mapping for Planetary Rovers from Terramechanics Perspective**, in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots Syst.*, Macau, China, China, 1869 - 1874 (2020). (Finalists of the **IROS ICROS Best Application Paper Award**)
- R. Zhou, W. Feng, H. Yang^{*}, H. Gao, N. Li, Z. Deng, L. Ding^{*}, **Predicting Terrain Mechanical Properties in Sight for Planetary Rovers with Semantic Clues**, *arXiv preprint*, arXiv:2011.01872, 2020.
- R. Zhou, W. Feng, Z. Deng, H. Gao, L. Ding^{*}, N. Li, **Sensitivity analysis and dominant parameter estimation of wheel-terrain interaction model**, *Acta Aeronautica et Astronautica Sinica*, 42(1), 24076 (2021). (in Chinese)
- L. Ding^{*}, P. Xu^{*}, H. Gao, Z. Wang, R. Zhou, Z. Gong, G. Liu, **Fault Tolerant Free Gait and Footstep Planning for Hexapod Robot Based on Monte-Carlo Tree**, *arXiv preprint*, arXiv:2006.07550, 2020.
- F. Lv, H. Gao, Y. Bai, N. Li, L. Ding, R. Zhou, Z. Deng, G. Liu, **Extraction of Speed-Independent Vibration Features for Terrain Classification in Lugged-Wheel Rovers**, in *Proc. IEEE Int. Conf. Robot. Biomimetics*, Kuala Lumpur, Malaysia, Malaysia, 1580 - 1585 (2018).

PROJECTS

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|---|------------------------------|
| ESA-ESRIC Space Resources Challenge <i>Python, Pytorch, ROS</i> | July. 2022 – September. 2022 |
| <ul style="list-style-type: none"> • Developed a rock segmentation model for legged robots working on emulated lunar environments. • Developed a scene semantic segmentation model for semantic mapping of legged robots working on emulated lunar environments. | |
| Common Terrain in Emulated Mars (CTEM) <i>Python, Pytorch</i> | Sep. 2019 – Nov. 2020 |
| <ul style="list-style-type: none"> • Established a well-annotated terrain semantic segmentation dataset for planetary scene understanding. • Developed a light-weighted terrain semantic segmentation model with competitive accuracy and speed for resources-limited rovers. | |

SCHOLARSHIP & AWARDS

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| ETH Robotics Research Fellowship | Jul. 2023 |
| National Scholarship for Graduate Students | Sep. 2022 |
| CSC Visiting Scholarship | Aug. 2021 |
| IROS ICROS Best Application Paper Award Finalist | Oct. 2019 |